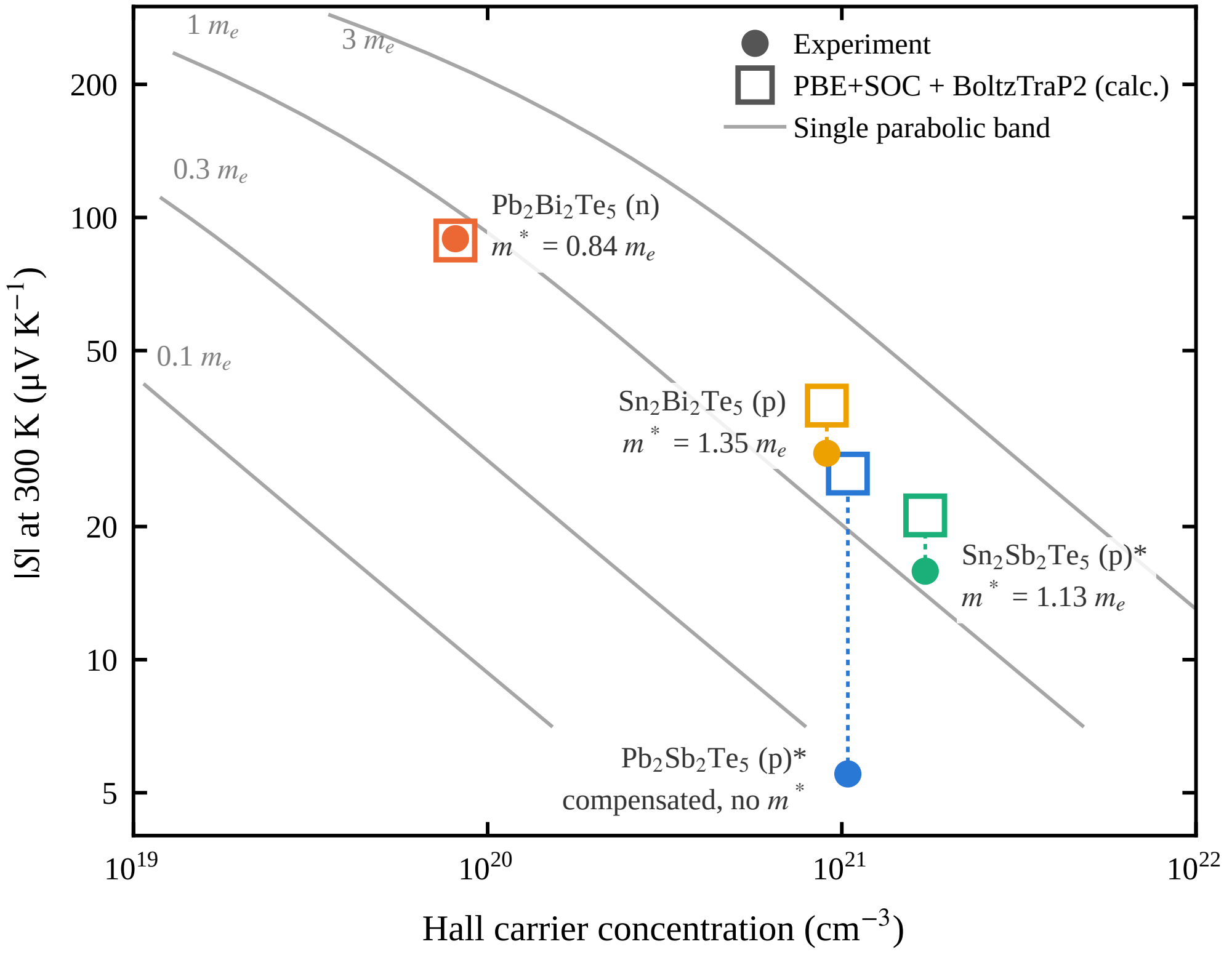
Experiment
PBE+SOC + BoltzTraP2 (calc.)
Single parabolic band
1 m_e
3 m_e
0.3 m_e
0.1 m_e
$Pb_2Bi_2Te_5$ (n)
m^* = 0.84 m_e
$Sn_2Bi_2Te_5$ (p)
m^* = 1.35 m_e
$Sn_2Sb_2Te_5$ (p)*
m^* = 1.13 m_e
$Pb_2Sb_2Te_5$ (p)*
compensated, no m^*
200
100
50
20
10
5
10^{19}
10^{20}
10^{21}
10^{22}
|S| at 300 K (μV K^{-1})
Hall carrier concentration (cm^{-3})